

数学物理方程 2 第三次作业 刘利士 2023511024

1. 1) $u_{xx} + 4u_{xy} + 5u_{yy} + u_x + 2u_y = 0$

$$\begin{cases} p = p(x, y) \\ q = q(x, y) \end{cases}$$

考虑特征线: $(\frac{dy}{dx})^2 - 4\frac{dy}{dx} + 5 = 0$

$$\Rightarrow \frac{dy}{dx} = 2 \pm i \Rightarrow y - (2 \pm i)x = C$$

$$\psi_1 = y - 2x$$

$$\psi_2 = x$$

$$\begin{cases} u_{xx} = \frac{\partial}{\partial x} (-2u_{\psi_1} + u_{\psi_2}) = 4u_{\psi_1\psi_1} - 4iu_{\psi_1\psi_2} + 2u_{\psi_2\psi_2} \\ u_{xy} = \frac{\partial}{\partial x} (u_{\psi_1}) = u_{\psi_1\psi_2} \\ u_{yy} = \frac{\partial}{\partial y} (u_{\psi_1}) = u_{\psi_1\psi_1} \\ u_x = (-2u_{\psi_1} + u_{\psi_2}) \\ u_y = u_{\psi_1} \end{cases}$$

$$\Rightarrow 4u_{\psi_1\psi_1} - 4u_{\psi_1\psi_2} - u_{\psi_2\psi_2} + 4u_{\psi_1\psi_2} + 5u_{\psi_1\psi_1} - 2u_{\psi_1} + u_{\psi_2} + 2u_{\psi_1} = 0$$

$$\Rightarrow 9u_{\psi_1\psi_1} + u_{\psi_2\psi_2} + u_{\psi_2} = 0$$

$$(2). u_{xx} + y u_{yy} + \frac{1}{2} u_y = 0 \quad \text{特征方程为: } \left(\frac{dy}{dx}\right)^2 + y = 0 \Rightarrow \frac{dy}{dx} = \pm \sqrt{y}$$

$$\Rightarrow 2\sqrt{y} \pm x = C. \quad \begin{cases} \psi_1 = 2\sqrt{y} \\ \psi_2 = x \end{cases}$$

$$\Rightarrow u_{\psi_2 \psi_2} + y \frac{\partial}{\partial y} \left(\frac{1}{\sqrt{y}} u_{\psi_1} \right) + \frac{1}{2\sqrt{y}} u_{\psi_1} = u_{\psi_2 \psi_2} + u_{\psi_1 \psi_1} = 0$$

$$(3) u_{xx} - 2\cos x u_{xy} - (3 + \sin^2 x) u_{yy} - y - u_y = 0$$

$$\text{特征方程为: } \left(\frac{dy}{dx}\right)^2 + 2\cos x \frac{dy}{dx} - (3 + \sin^2 x) = 0$$

$$\Rightarrow \frac{dy}{dx} = -\cos x \pm 2 \Rightarrow y + \sin x \pm 2x = C$$

$$\begin{cases} \psi_1 = y + \sin x \\ \psi_2 = 2x \end{cases}$$

$$\Rightarrow \begin{cases} u_{xx} = \frac{\partial}{\partial x} (\cos x u_{\psi_1} + 2 u_{\psi_2}) = -\sin x u_{\psi_1} + \cos^2 x u_{\psi_1 \psi_1} \\ u_{xy} = 2 u_{\psi_1 \psi_2} \\ u_{yy} = u_{\psi_1 \psi_1} \\ u_y = u_{\psi_1} \end{cases} + 4\cos x u_{\psi_1 \psi_2} + 4 u_{\psi_2 \psi_2}$$

$$\Rightarrow \text{原方程变为: } (\cos^2 x - 3 - \sin^2 x) u_{\psi_1 \psi_1} + 4 u_{\psi_2 \psi_2} - \sin x u_{\psi_1} - \psi_1 + \sin x - u_{\psi_1} = 0$$

$$2. (1) a_1(x) u_{xx} + b_1(y) u_{yy} + a_2(x) u_x + b_2(y) u_y = 0$$

令 $u = X(x) Y(y)$, 则原方程变为:

$$a_1 X_{xx} Y + b_1 X Y_{yy} + a_2 X_x Y + b_2 X Y_y = 0$$

$$\Rightarrow \begin{cases} a_1 X_{xx} + a_2 X_x = \lambda X \\ b_1 Y_{yy} + b_2 Y_y = -\lambda Y \end{cases}$$

(2) $\frac{1}{r} \frac{\partial}{\partial r} (r u_r) + \frac{1}{r^2} u_{\varphi\varphi} = 0$. 令 $u = R(r) \Phi(\varphi)$. 则原方程变为:

$$R_{rr} \Phi + \frac{1}{r} R_r \Phi + \frac{1}{r^2} R \Phi_{\varphi\varphi} = 0$$

$$\Rightarrow \frac{\Phi_{\varphi\varphi}}{\Phi} + \frac{r^2}{R} R_{rr} + \frac{r}{R} R_r = 0 \Rightarrow \begin{cases} \Phi_{\varphi\varphi} = \lambda \Phi \\ r^2 R_{rr} + r R_r = -\lambda R \end{cases}$$

$$(3) \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial u}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial u}{\partial \theta}) = 0 \quad \text{设 } u = R(r) \Theta(\theta).$$

$$\Rightarrow \frac{2}{r} R_r \Theta + R_{rr} \Theta + \frac{1}{r^2} R \Theta_{\theta\theta} + \frac{1}{r^2} \cot \theta R \Theta_{\theta} = 0$$

$$\Rightarrow \begin{cases} 2r R_r + r^2 R_{rr} = \lambda R \\ \Theta_{\theta\theta} + \cot \theta \Theta_{\theta} = -\lambda \Theta \end{cases}$$

$$3. \begin{cases} u_{tt} = a^2 u_{xx} \quad (0 < x < L, t > 0) \\ u_x|_{x=0} = 0, \quad u_x|_{x=L} = 0 \\ u|_{t=0} = e^{-x^2}, \quad u_t|_{t=0} = 2ax e^{-x^2} \end{cases}$$

该齐次边界条件的基为 $\cos(\frac{n\pi}{L} x)$ ($n=0, 1, 2, \dots$)

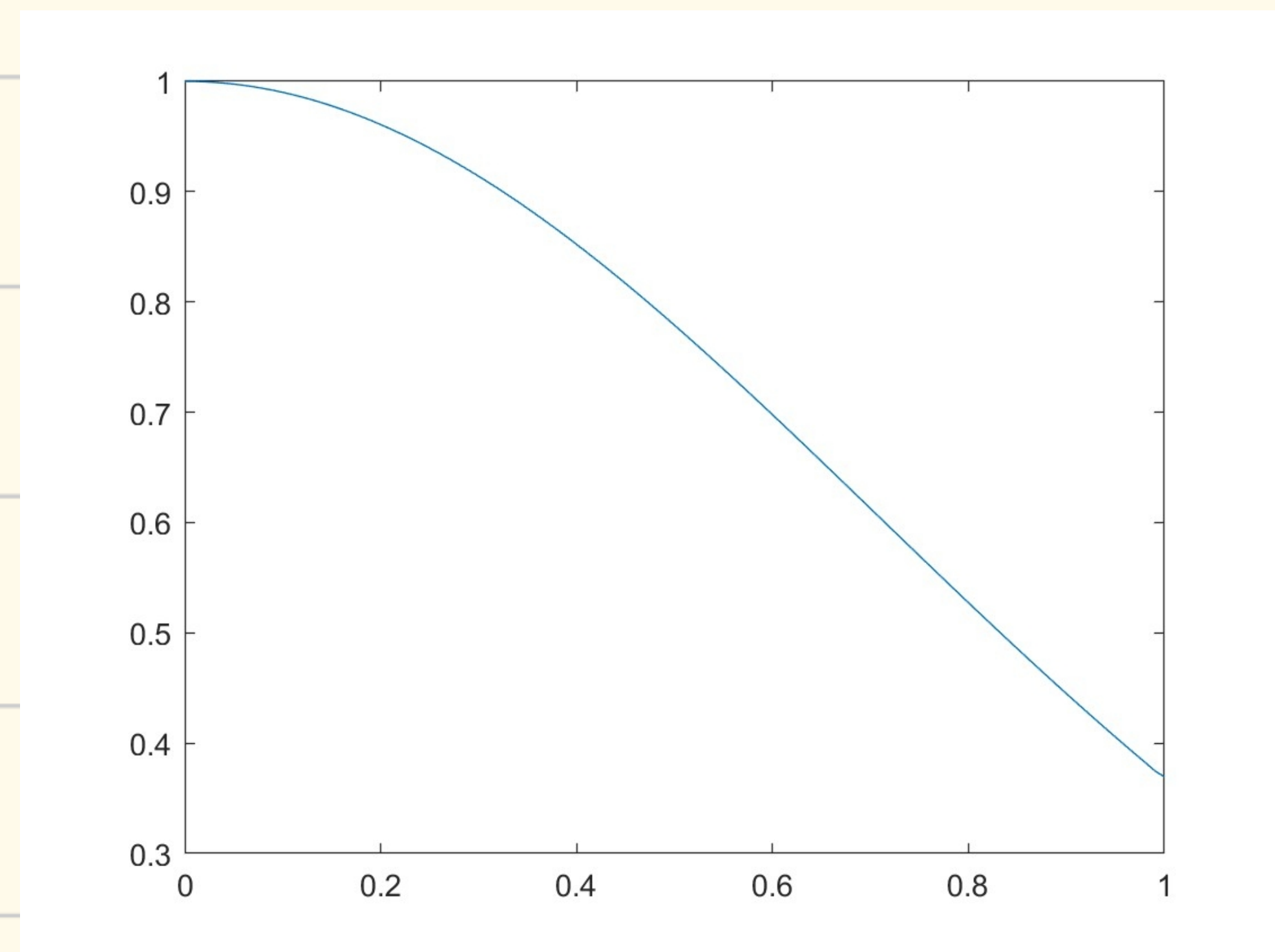
$$\text{设 } u(x, t) = \sum_{n=0}^{\infty} (A_n \cos(a \frac{n\pi}{L} t) + B_n \sin(a \frac{n\pi}{L} t)) \cdot \cos(\frac{n\pi}{L} x).$$

由初始条件及基函数的正交性, 我们有

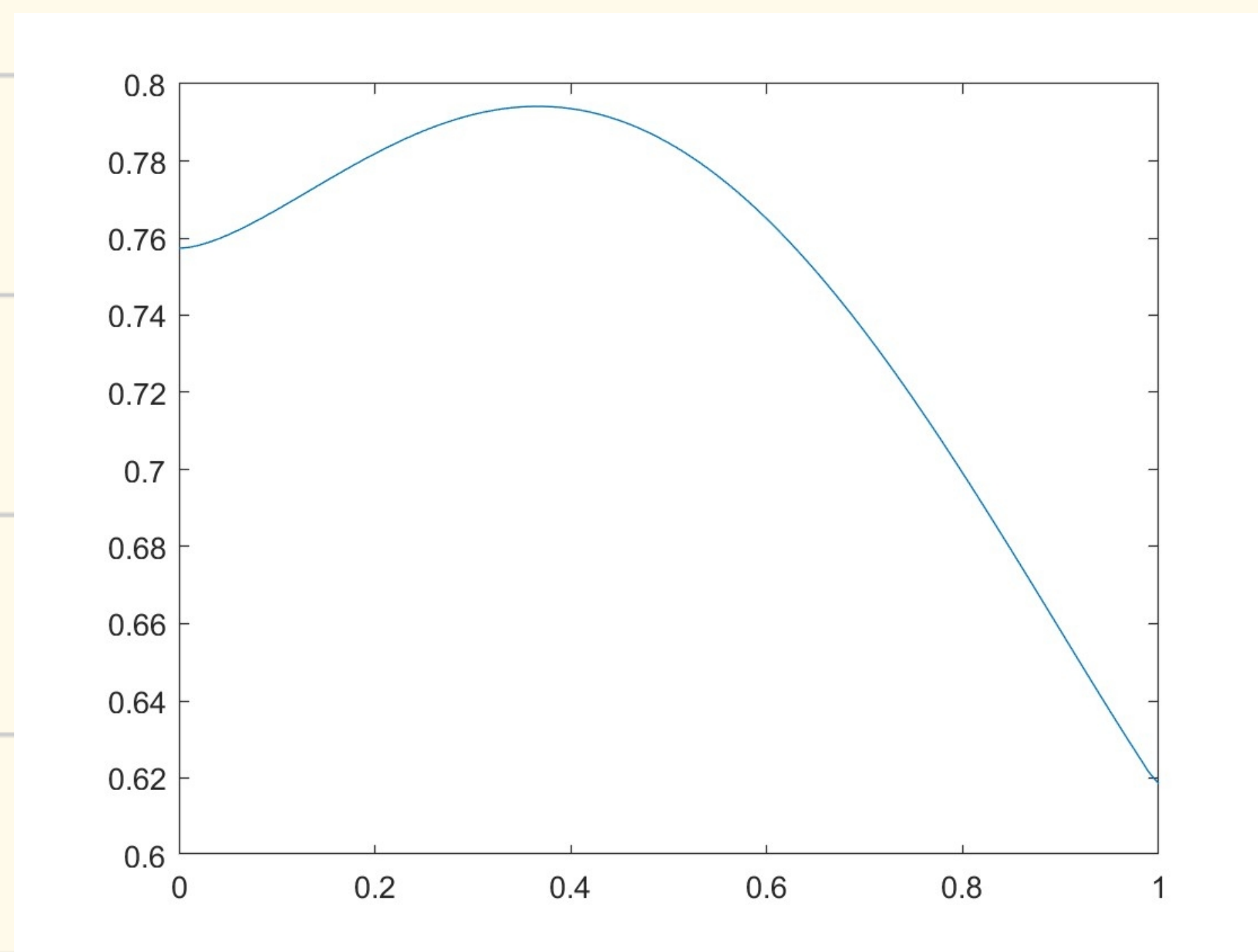
$$\begin{cases} A_n = \frac{2}{L} \int_0^L e^{-x^2} \cos(\frac{n\pi}{L} x) dx \\ B_n = \frac{2}{L} \int_0^L \frac{2}{an\pi} \cdot 2ax e^{-x^2} \cos(\frac{n\pi}{L} x) dx \end{cases} \quad n=1, 2, \dots$$

$A_0 = \frac{1}{L} \int_0^L e^{-x^2} dx$. A_n, B_n 无初等函数解

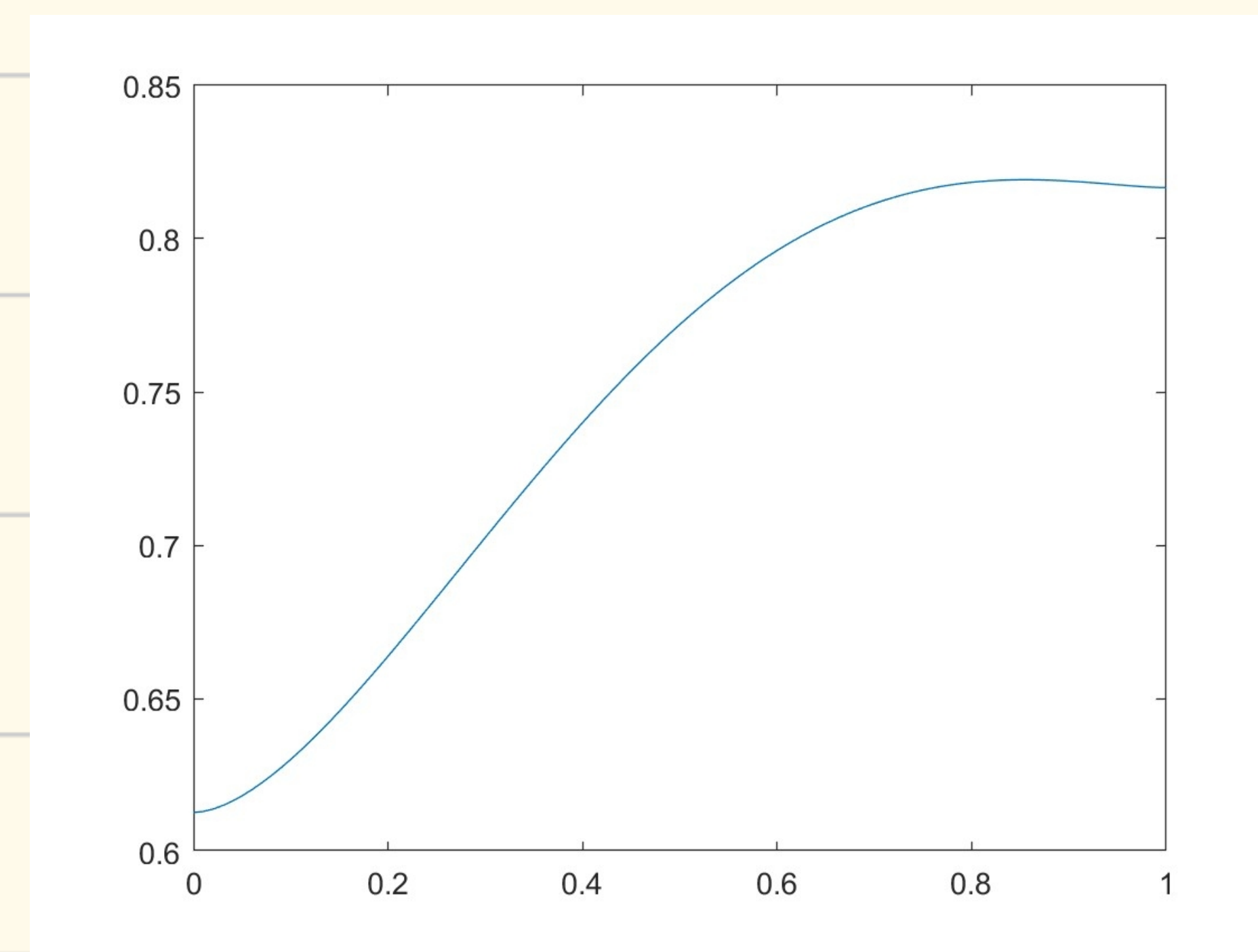
考虑 $a=1, L=1$ 时的数值解 取 $n=100$ 截断. 此时可以得到波形图:



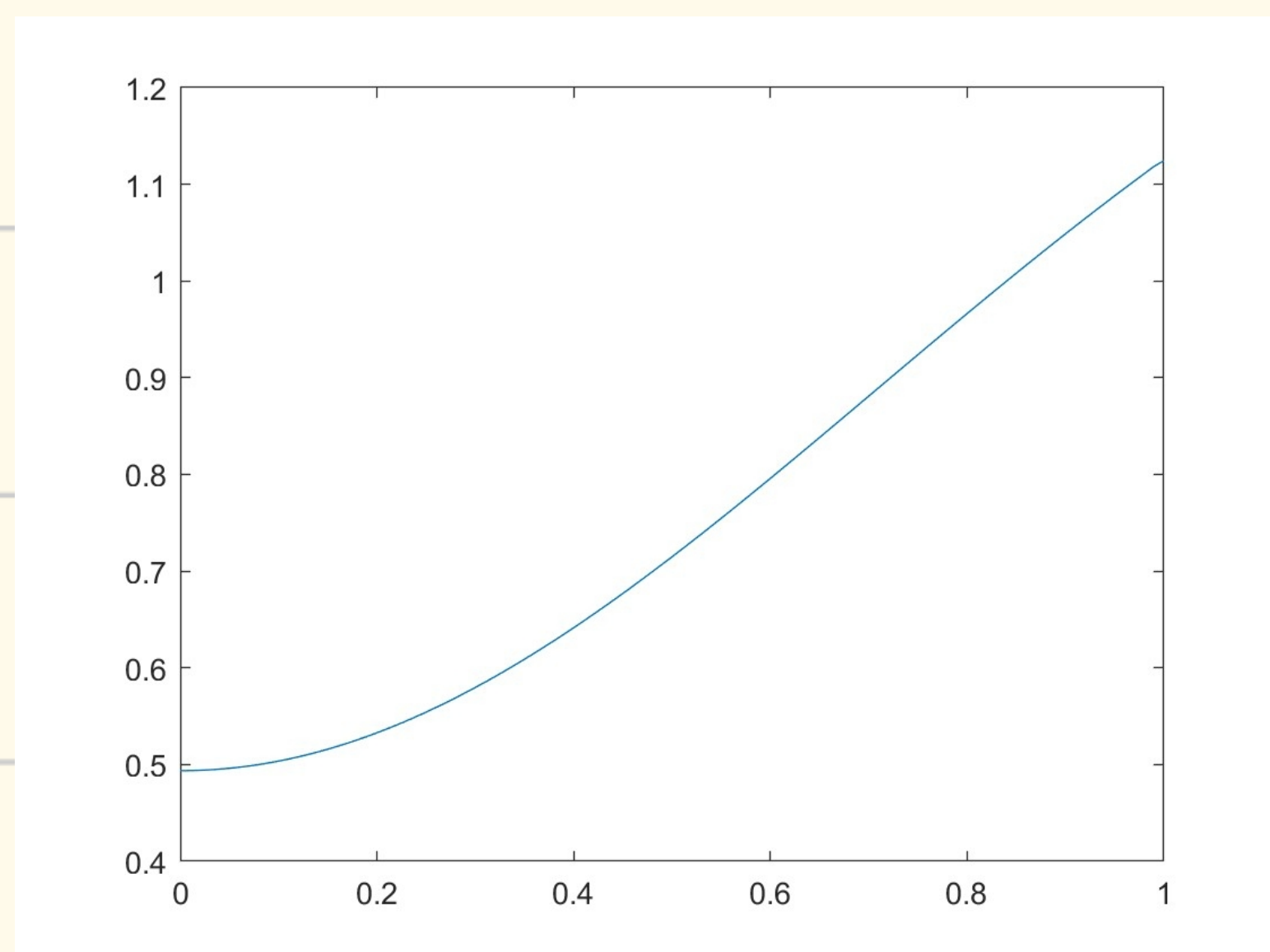
$t=0$



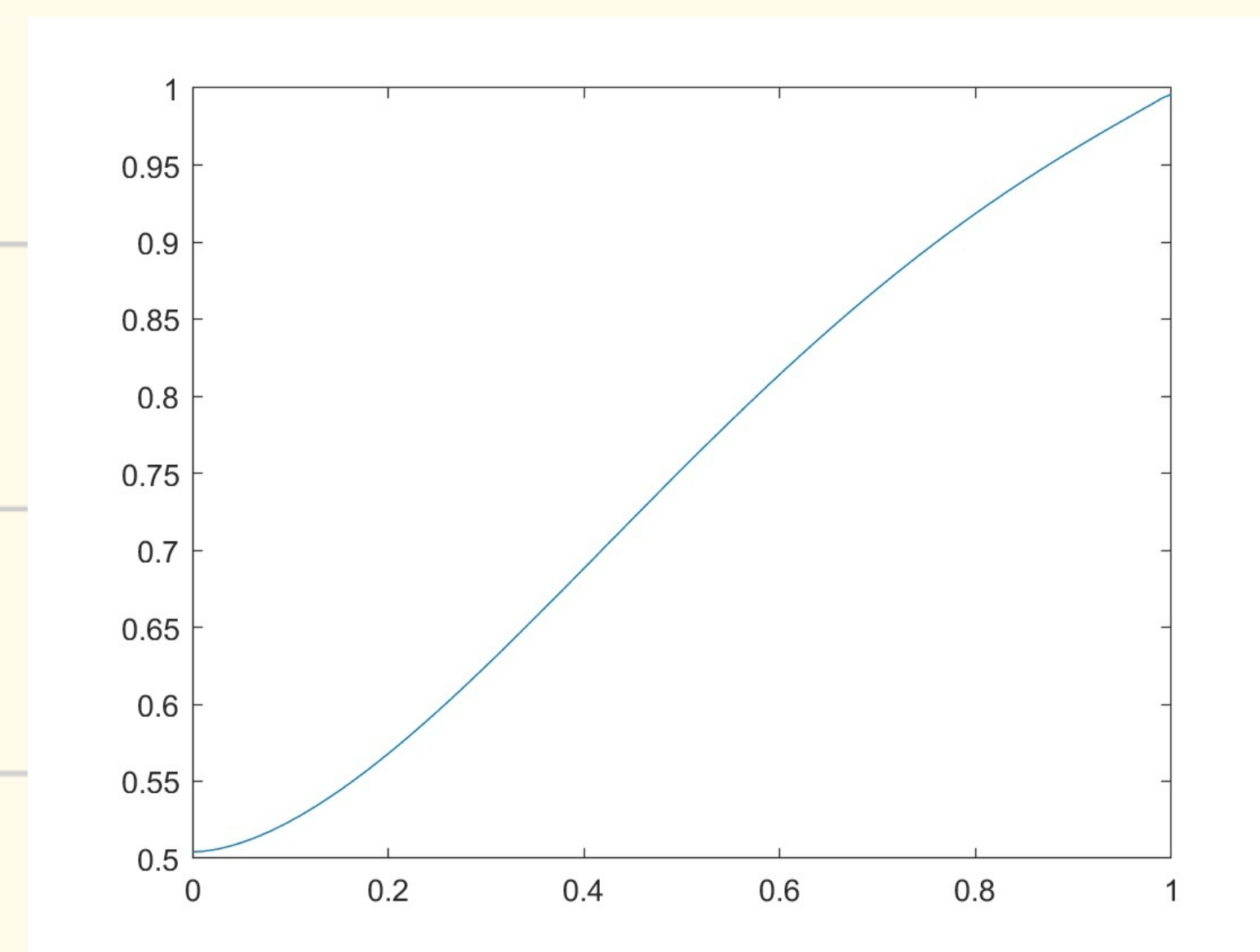
$t=1/3$



$t=1/2$



$t=2/3$



$t=1$

$$4. \begin{cases} u_t = a^2 u_{xx} & (0 < x < L, t > 0) \\ u|_{x=0} = 0, \quad u|_{x=L} = 0 \\ u|_{t=0} = b x(1-x)/L^2 \end{cases}$$

由齐次边界条件的基, 设 $u = \sum_{n=1}^{+\infty} A_n e^{-a^2 (\frac{n\pi}{L})^2 t} \sin(\frac{n\pi}{L} x)$

由初始条件以及基函数的正交性, 我们有:

$$A_n = \frac{2}{L} \int_0^L \frac{b}{L^2} x(1-x) \sin(\frac{n\pi}{L} x) dx = \frac{2b(2 - 2\cos(n\pi) - n\pi \sin(n\pi))}{n^3 \pi^3}$$

因此解为

$$u(x, t) = \sum_{n=1}^{+\infty} \frac{2b(2 - 2\cos(n\pi) - n\pi \sin(n\pi))}{n^3 \pi^3} e^{-a^2 (\frac{n\pi}{L})^2 t} \sin(\frac{n\pi}{L} x)$$

5. 由泊松公式,

$$u(\vec{x}, t) = \frac{\partial}{\partial t} \left(\frac{1}{4\pi a^2 t} \oint_{\partial B(\vec{x}, at) \cap B(\vec{0}, R)} u_0 \cdot d\vec{s} \right) + \frac{1}{4\pi a^2 t} \oint_{\partial B(\vec{x}, at)} 0 \cdot d\vec{s},$$

设 $\vec{x} = (x, y, z)$ $r_0 = \sqrt{x^2 + y^2 + z^2}$. 当 $r_0 \geq at + R$ 或 $r_0 + R \leq at$ 时, $u(\vec{x}, t) = 0$.

当 $r_0 - R \leq at \leq r_0 + R$ 时.

$$\oint_{\partial B(\vec{x}, at) \cap B(\vec{0}, R)} u_0 d\vec{s} = \int_0^{2\pi} d\varphi \int_0^{\arccos \frac{R^2 - a^2 t^2 - r_0^2}{2at r_0}} u_0 \sin \theta a^2 t^2 d\theta$$

$$= u_0 a^2 t^2 \cdot 2\pi \left(1 - \frac{R^2 - a^2 t^2 - r_0^2}{2at r_0} \right) = 2\pi u_0 \left(a^2 t^2 - \frac{at}{2r_0} (R^2 - a^2 t^2 - r_0^2) \right)$$

$$\Rightarrow u(\vec{x}, t) = \frac{\partial}{\partial t} \left(\frac{u_0}{2a^2} \left(a^2 t - \frac{a}{2r_0} (R^2 - a^2 t^2 - r_0^2) \right) \right)$$
$$= \frac{1}{2} u_0 + u_0 \frac{at}{2r_0}.$$

$$\text{因此 } u(\vec{x}, t) = \begin{cases} 0 & \|\vec{x}\| - R \geq at \text{ 或 } \|\vec{x}\| + R \leq at \\ \frac{1}{2} u_0 + u_0 \frac{at}{\|\vec{x}\|} \frac{1}{2} & \|\vec{x}\| - R \leq at \leq \|\vec{x}\| + R \end{cases}$$

$$6. \begin{cases} u_{xx} + u_{yy} = 6(x+y)^2 & 0 < a^2 < x^2 + y^2 < b^2 \\ u|_{x^2+y^2=a^2} = 1, \quad \frac{\partial u}{\partial n}|_{x^2+y^2=b^2} = 0 \end{cases}$$

考虑换元 $\begin{cases} x = r \cos \theta \\ y = r \sin \theta. \end{cases} \Rightarrow \frac{1}{r} \frac{\partial}{\partial r} (r u_r) + \frac{1}{r^2} u_{\theta\theta} = 6r^2 (\cos \theta + \sin \theta)^2 = 6r^2 (\sin 2\theta + 1)$

由于周期性边界条件 $u(r, \theta) = u(r, \theta + 2\pi)$ u 可展开为:

$$u = \sum_{n=0}^{\infty} (A_n(r) \cos n\theta + B_n(r) \sin n\theta). \text{ 由非齐次方程和基矢函数的正交性,}$$

我们有:
$$\begin{cases} \frac{d^2 A_0}{dr^2} + \frac{1}{r} \frac{dA_0}{dr} = 6r^2 \\ \frac{d^2 B_2}{dr^2} + \frac{1}{r} \frac{dB_2}{dr} - \frac{4}{r^2} B_2 = 6r^2 \\ A_n = 0 \quad \forall n \neq 0 \\ B_n = 0 \quad \forall n \neq 2 \end{cases}$$

考虑欧拉方程化换: $\eta = \ln r \Rightarrow \frac{d^2 A_0}{dr^2} = \frac{d}{dr} \left(\frac{dA_0}{d\eta} \cdot \frac{1}{r} \right) = \frac{1}{r^2} \frac{d^2 A_0}{d\eta^2} - \frac{1}{r^2} \frac{dA_0}{d\eta}$

$$\frac{dA_0}{dr} = \frac{1}{r} \frac{dA_0}{d\eta}$$

$$\Rightarrow \frac{d^2 A_0}{d\eta^2} = 6r^4 = 6e^{4\eta} \Rightarrow A_0 = \frac{3}{8}e^{4\eta} + C_1\eta + C_2 = \frac{3}{8}r^4 + C_1 \ln r + C_2$$

同理可得: $\frac{d^2 B_2}{d\eta^2} - 4B_2 = 6e^{4\eta}$ 考虑 Fourier 变换.

$$-\omega^2 \hat{B}_2 - 4\hat{B}_2 = 12\pi \delta(4) \Rightarrow B_2 = C_3 e^{2\eta} + C_4 e^{-2\eta} + \frac{1}{2}e^{4\eta}.$$

$$\text{因此得解: } u = \frac{3}{8}r^4 + C_1 \ln r + C_2 + \left(\frac{1}{2}r^4 + C_3 r^2 + \frac{C_4}{r^2}\right) \sin 2\theta$$

$$\text{由边界条件: } \begin{cases} \frac{3}{8}a^4 + C_1 \ln a + C_2 = 1 \\ \frac{1}{2}a^4 + C_3 a^2 + C_4/a^2 = 0 \\ \frac{3}{2}b^4 + C_1/b = 0 \\ 2b^3 + 2C_3 b - 2C_4/b^3 = 0 \end{cases} \Rightarrow$$

$$\begin{cases} C_1 = \frac{3}{2}b^5 \\ C_2 = 1 - \frac{3}{8}a^4 - \frac{3}{2}b^5 \ln a \\ C_3 = \frac{-a^6 - \frac{1}{2}b^6}{2(a^4 + b^4)} \\ C_4 = \frac{-a^4 b^6}{2(a^4 + b^4)} \end{cases}$$

$$\text{综上, } u = \frac{3}{8} r^4 + \frac{3}{2} b^5 \ln r + 1 - \frac{3}{8} a^4 - \frac{3}{2} b^5 \ln a + \left(\frac{1}{2} r^4 - \frac{2ab + b^6}{4(a^4 + b^4)} r^2 - \frac{a^4 b^6}{4(a^4 + b^4)} \cdot \frac{1}{r^2} \right) \sin 2\theta.$$

$$7. \text{ 设 } u = \sum_{n=0}^{+\infty} \left(A_n \cos\left(\frac{a}{L}\left(n+\frac{1}{2}\right)\pi t\right) + B_n \sin\left(\frac{a}{L}\left(n+\frac{1}{2}\right)\pi t\right) \right) \sin\left(\frac{x}{L}\left(n+\frac{1}{2}\right)\pi\right)$$

(1) 显然有 $B_n = 0$ (由初始条件)

$$A_n = \frac{\int_0^L \sin x \sin\left(\frac{1}{L}\left(n+\frac{1}{2}\right)\pi x\right) dx}{\int_0^L \sin^2\left(\frac{1}{L}\left(n+\frac{1}{2}\right)\pi x\right) dx} = \frac{2}{L} \int_0^L \sin x \sin\left(\frac{x}{L}\left(n+\frac{1}{2}\right)\pi\right) dx$$

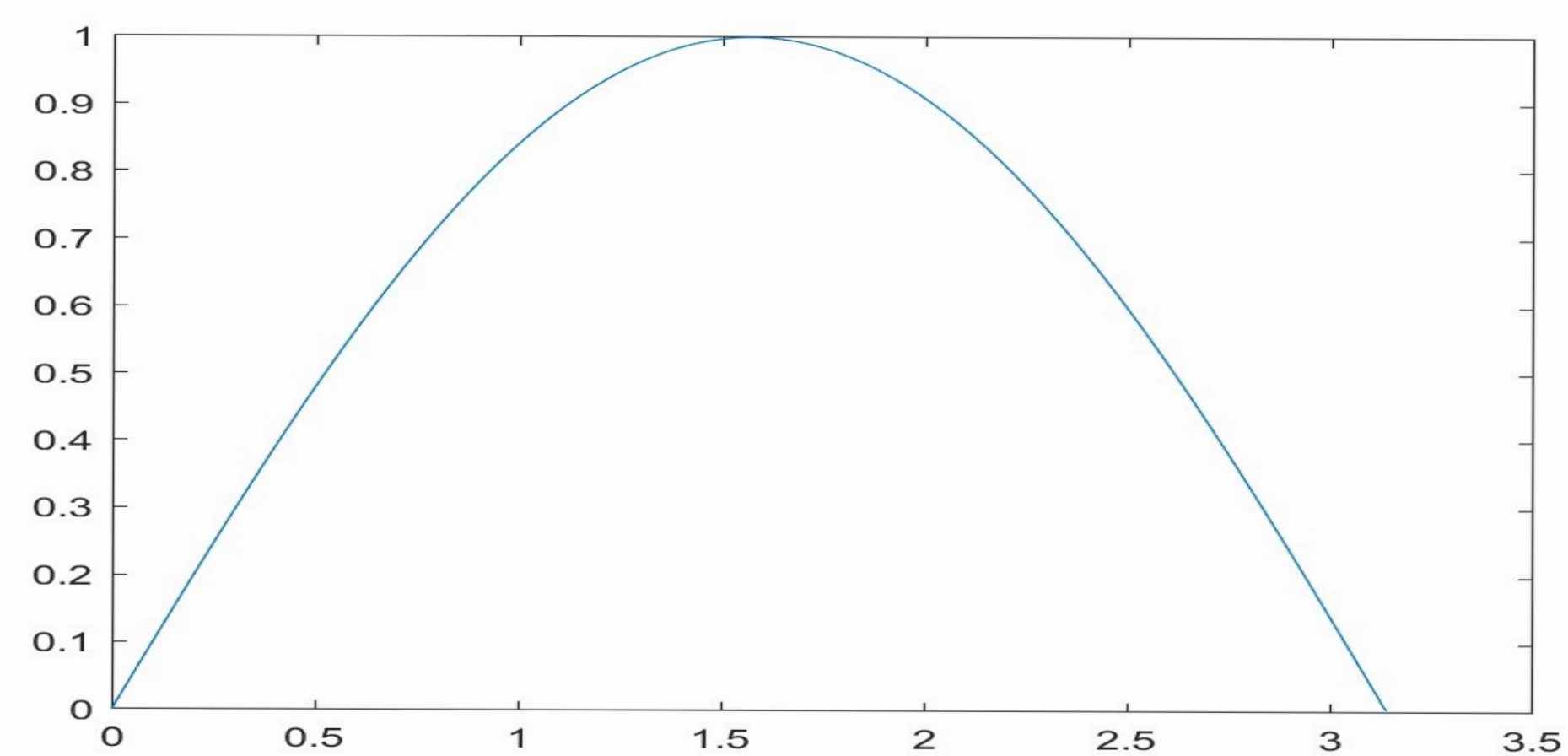
$$= \frac{2}{L} \int_0^L \frac{1}{2} \left(\cos\left(x\left(\frac{n+\frac{1}{2}}{L}\pi - 1\right)\right) - \cos\left(x\left(\frac{n+\frac{1}{2}}{L}\pi + 1\right)\right) \right) dx$$

$$= \frac{1}{L} \left[\frac{L}{(n+\frac{1}{2})\pi - L} \sin\left((n+\frac{1}{2})\pi - L\right) - \frac{L}{(n+\frac{1}{2})\pi + L} \sin\left((n+\frac{1}{2})\pi + L\right) \right]$$

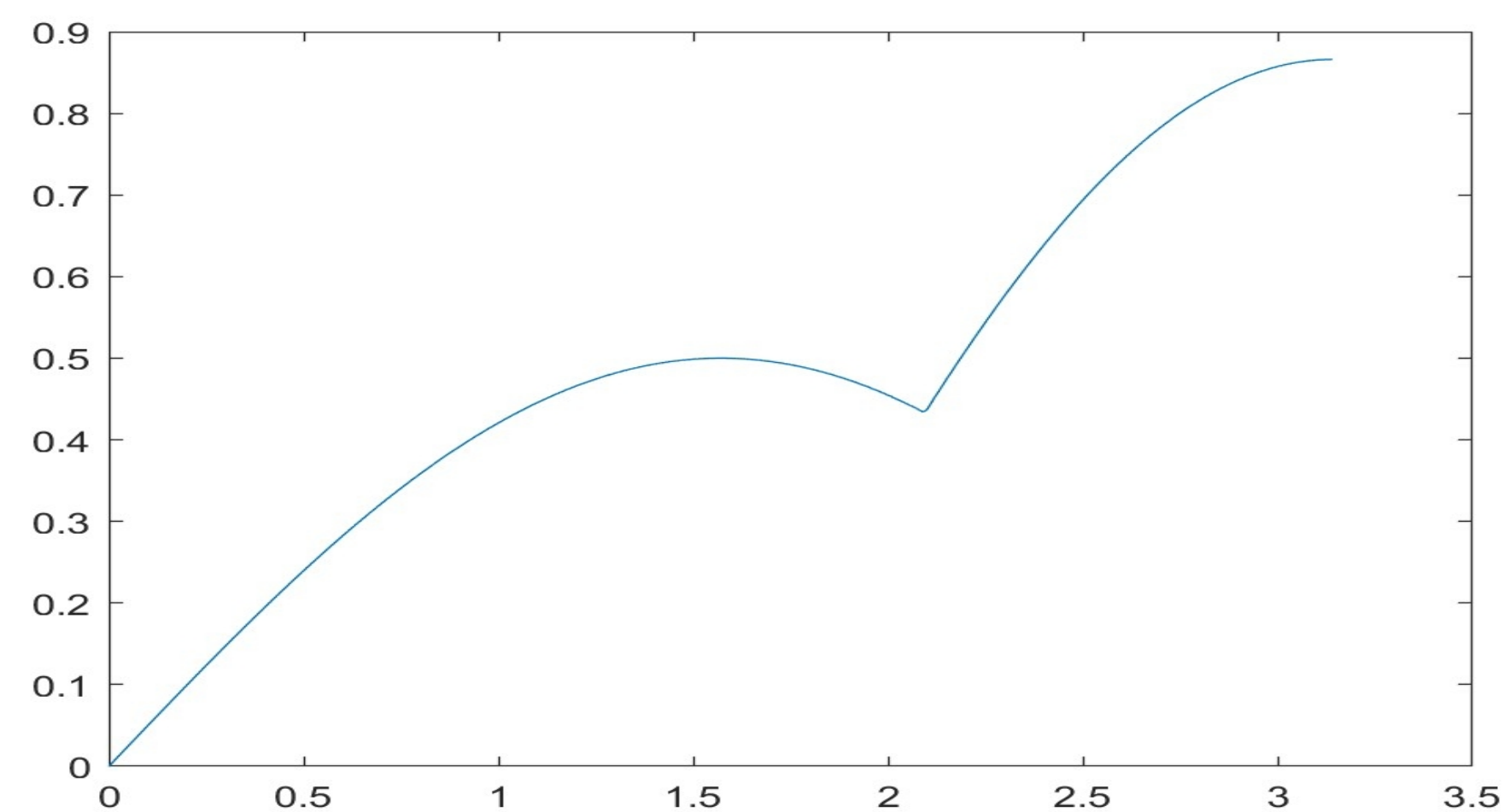
$$= (-1)^n \cos L \cdot \frac{2L}{(n+\frac{1}{2})^2 \pi^2 - L^2}$$

因此解为 $u(x,t) = \sum_{n=0}^{+\infty} (-1)^n \frac{2L \cos L}{(n+\frac{1}{2})^2 \pi^2 - L^2} \cos(\frac{a}{L}(n+\frac{1}{2})\pi t) \sin(\frac{1}{L}(n+\frac{1}{2})\pi x)$

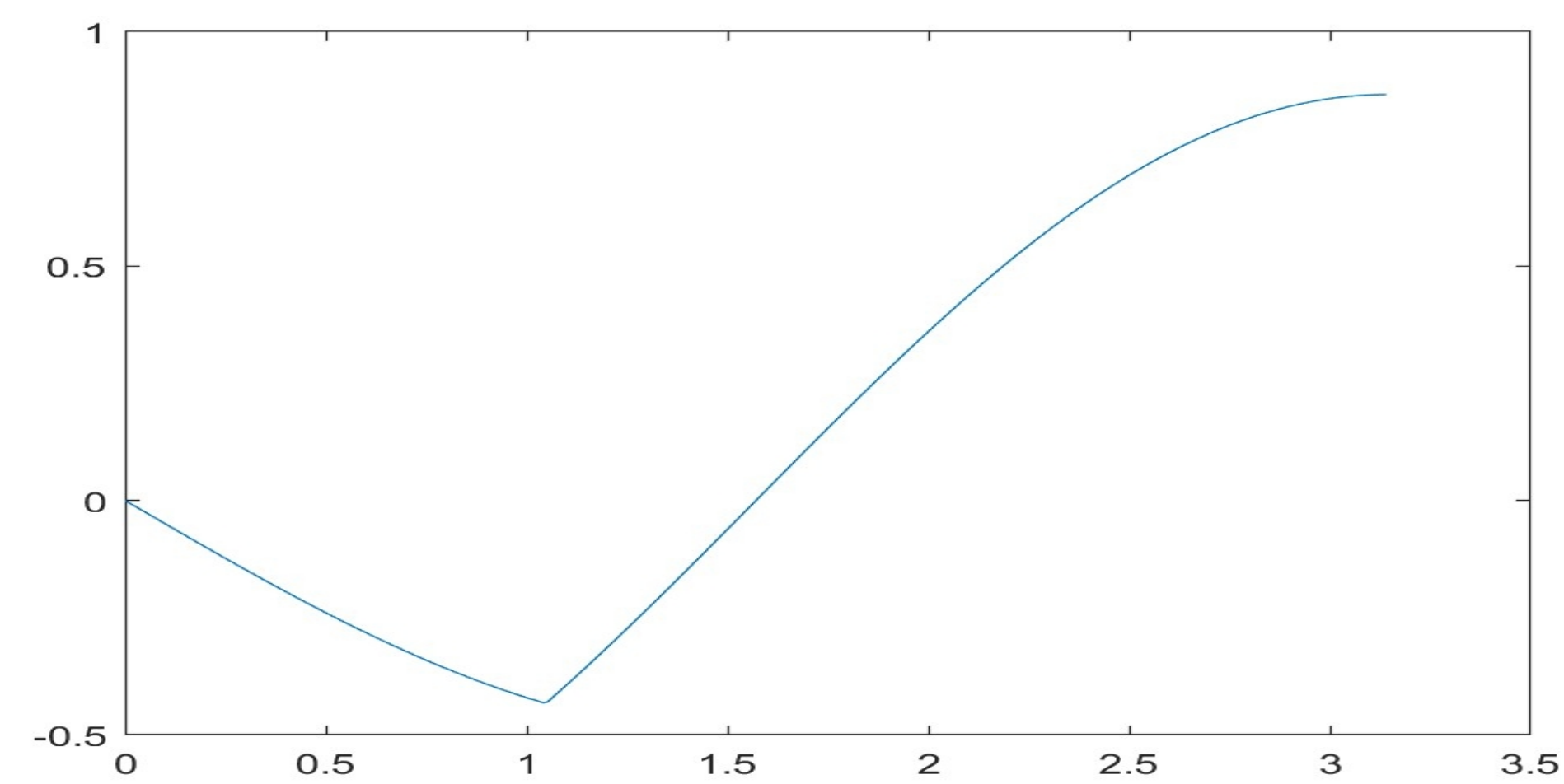
考虑 $L = \pi$ 时的振动情况: (视频页可见附件 2)
($a=1$)



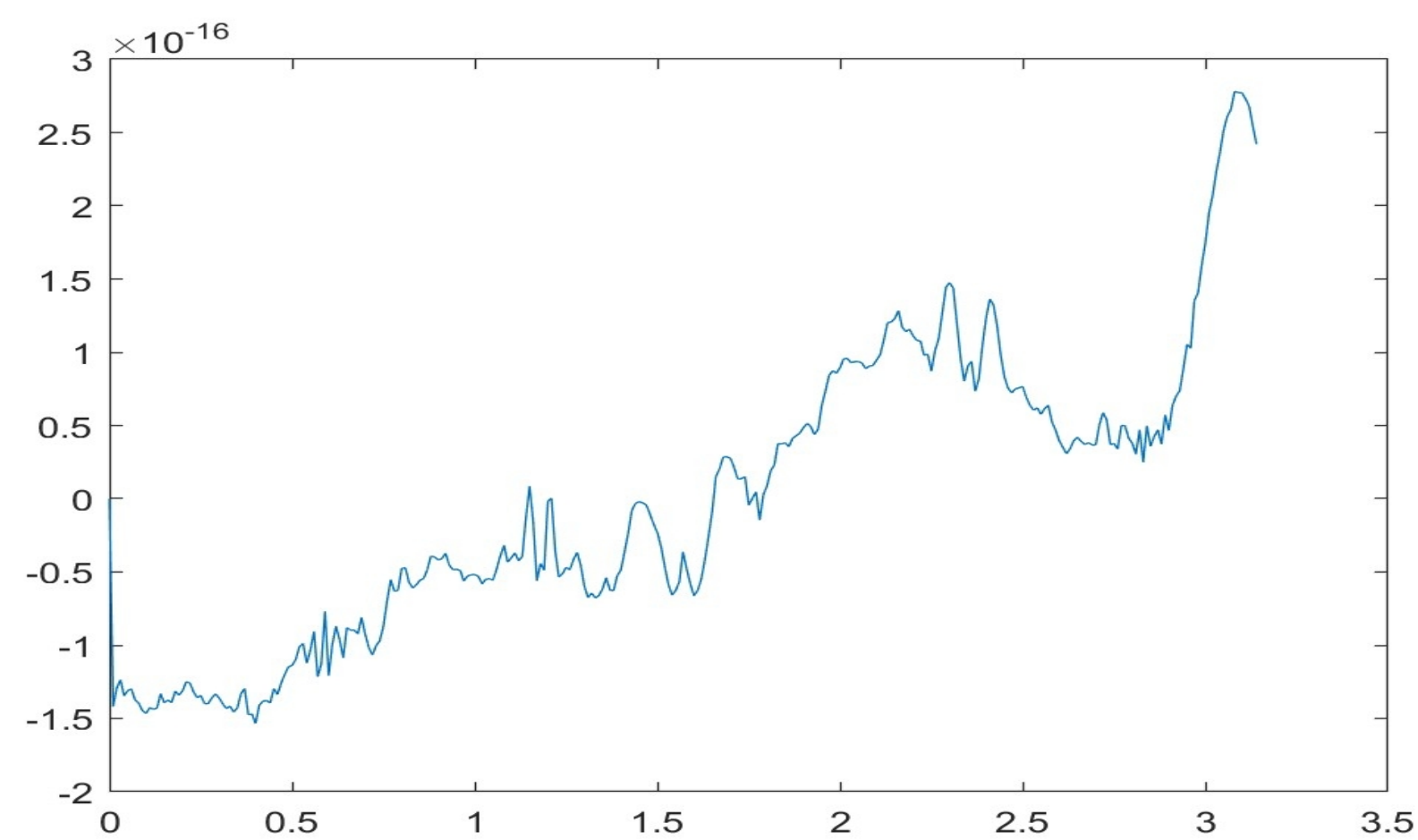
$t=0$



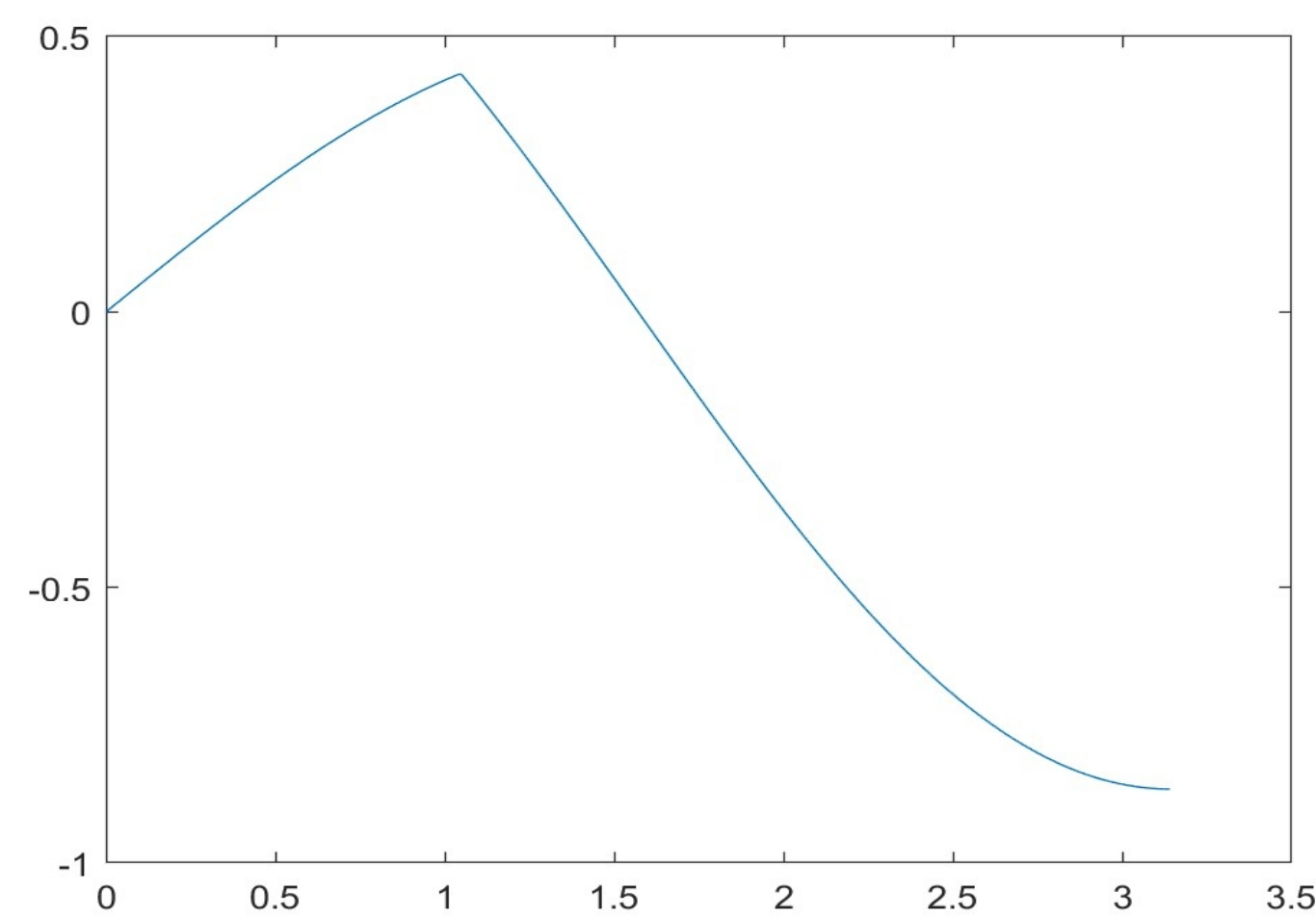
$t = \frac{1}{3}\pi$



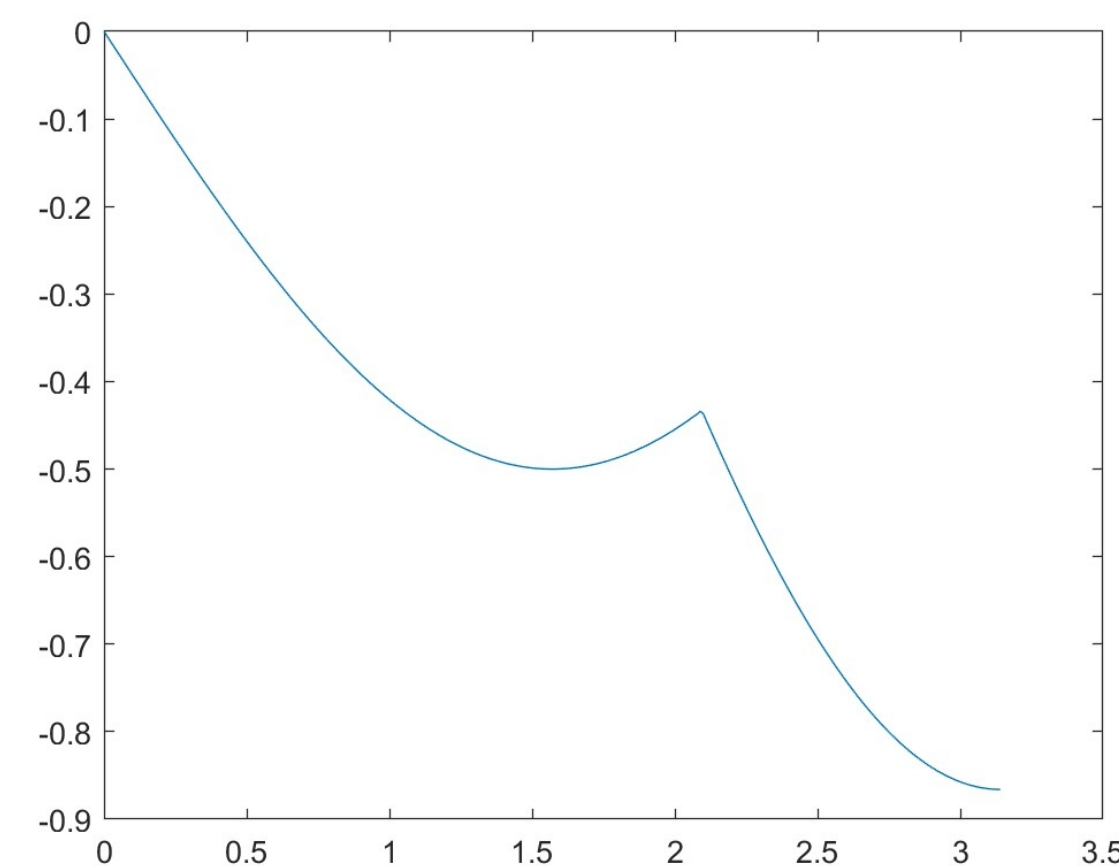
$t = \frac{2}{3}\pi$



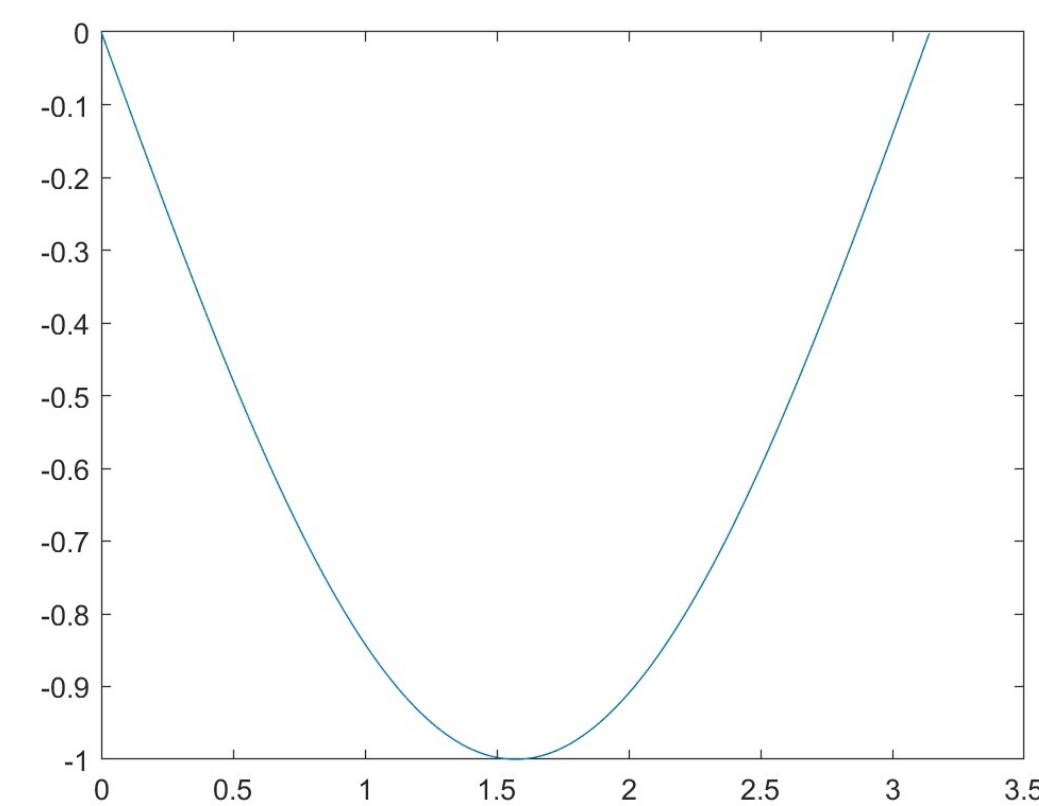
$t=\pi$



$t = \frac{4}{3}\pi$



$t = \frac{5}{3}\pi$



$t=2\pi$

(2) 当 $L \rightarrow +\infty$ 且 $\frac{t}{L} \ll 1$ 时

$$u = \sum_{n=0}^{+\infty} (-1)^n \frac{2L \cos L}{(n + \frac{1}{2})^2 \pi^2 - L^2} \cos\left(\frac{a}{L}(n + \frac{1}{2})\pi t\right) \sin\left(\frac{1}{L}(n + \frac{1}{2})\pi x\right)$$

$\sim \cos(at) \sin x$? 不会